

Investigating Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER)

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Abstract

Introduction & Literature Review

- Postoperative hypoxemia is a frequent complication from anesthetic¹
- The MOVER EPIC dataset includes records from 64,354 surgeries between 2017 and 2022² and can be used to study this issue
- Prior research has assessed how postoperative hypoxemia is affected by different variables, but these models only contained up to four laboratory test results^{3,4,5}
- To fill this gap characterized by few laboratory test results, this work modelled postoperative hypoxemia probability using logistic regression with up to ten preoperative laboratory test results and three demographic variables as covariates
- The objective was to detect significant predictors of postoperative hypoxemia and determine the accuracy of predicting its probability with just these variables

Methodology (The Pipeline)

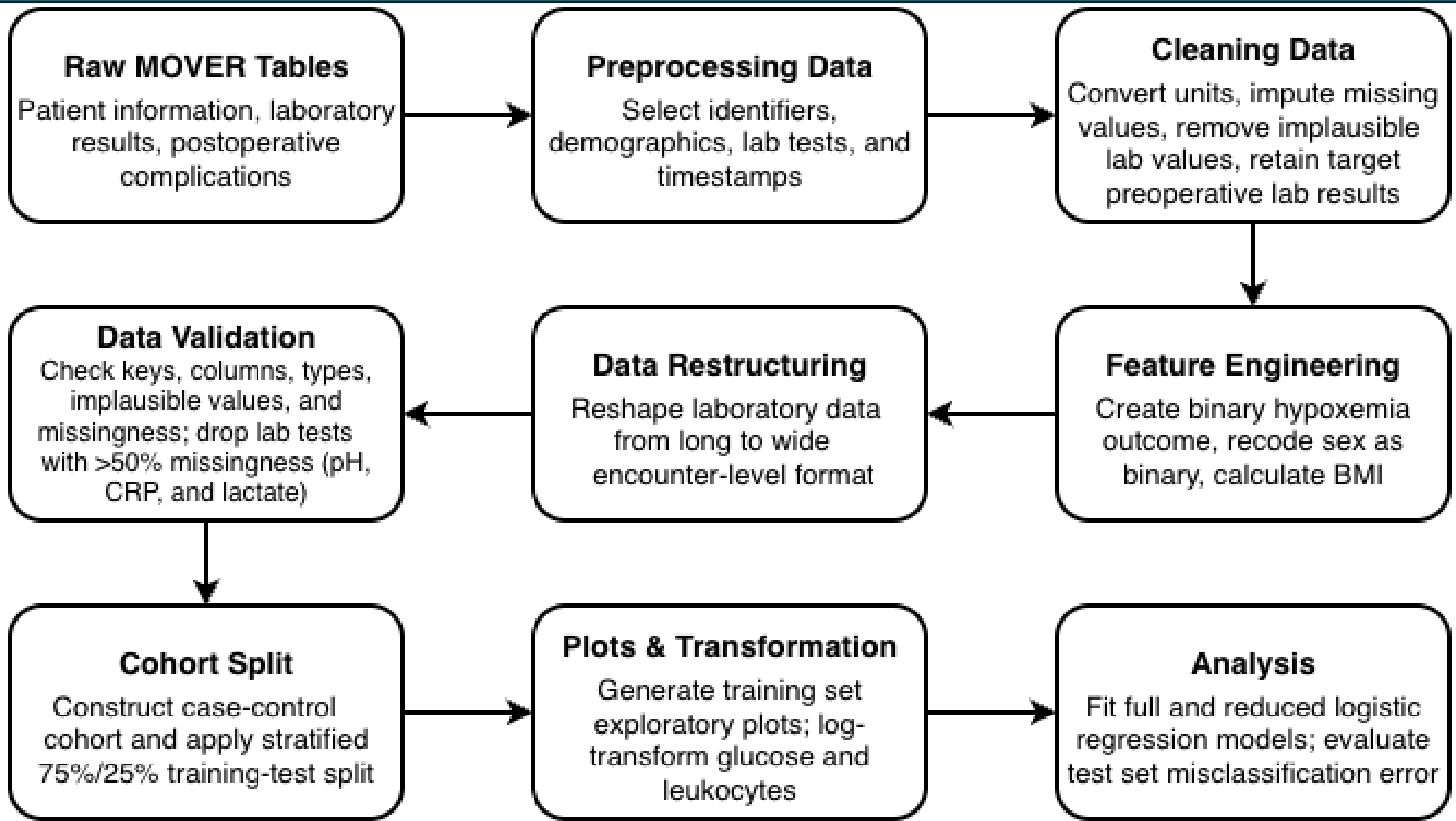


Figure 1. Flow chart of data wrangling architecture

Key Findings

Variable	Coefficient Estimate	P-value
Intercept	-6.576	0.237
Age	0.004	0.660
Sex: Female	-1.426	0.001
Body mass index	0.051	0.040
Carbon dioxide	0.015	0.716
log(Glucose)	0.786	0.114
Hematocrit	0.094	0.557
Hemoglobin	-0.521	0.264
log(Leukocytes)	0.257	0.485
Potassium	0.117	0.717
Sodium	0.014	0.650

Table 1. Full logistic regression model results for postoperative hypoxemia

- On the test set, the full model had a misclassification error rate of 0.293 (29.3%), while the reduced model, which included only the predictors that were significant in the full model, had a misclassification error rate of 0.28 (28.0%). Thus, the reduced model showed slightly better predictive performance than the full model in this sample.

- Table 1 presents the full logistic regression model results. In the full model, sex and body mass index were the only predictors significant at the 0.05 level, with female sex associated with lower odds of postoperative hypoxemia and higher body mass index associated with higher odds.

Conclusion & Impact

Key Takeaway: Patients who are **male** or **have high BMI** may benefit from closer postoperative monitoring for timely detection and intervention of hypoxemia

Key Risk Factors

- **Higher BMI** elevates odds of postoperative hypoxemia, aligning with established literature⁴
- **Male** patients have higher odds of hypoxemia, which contradicts previous findings that showed being female as a risk factor⁴

Model Performance and Limitation

- The reduced model achieved a lower test set misclassification rate (28%) than the full model (29%), suggesting that additional demographic and lab test variables introduced noise rather than clinical signal
- Only 98 out of 24,167 patients (0.4%) had hypoxemia. This extreme class imbalance remains a challenge for clinical application, as the high misclassification rate persisted despite undersampling control cases to balance the data

Future Directions

- Employing systematic oversampling or re-weighting techniques to improve minority class classification
- Exploring alternative biomarkers, lab tests and the interplay between hypoxemia and other concurrent postoperative complications to enhance predictive accuracy

References

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